

SAPIENZA
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Network Optimization

Volume 3 — Flows

Network flow models and total unimodularity; maximum flow and minimum cut; minimum cost flow with negative cycles, potentials and reduced costs; transportation, transshipment and assignment problems.

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Chapter 4

Network flow models

Prerequisites

- From this module, Chapter 1 «Graphs»: directed graphs, the node–arc incidence matrix, spanning trees and the rank of the incidence matrix.
- From the module Linear Programming, Chapter 9 «Duality theory» and Chapter 15 «The geometry of Linear Programming»: the dual of an LP, weak and strong duality, vertices of a polyhedron.
- Determinants: the Laplace expansion along a column and Cramer's rule.

Objectives

At the end of this chapter you will be able to:

- describe a network with balances, lower bounds, capacities and costs, and write the minimum cost flow model, also in matrix form with the incidence matrix;
- check the necessary conditions for the existence of a feasible flow, and build counterexamples;
- remove the lower bounds by a substitution, and balance an instance with a dummy node;
- prove that the incidence matrix is totally unimodular, and deduce that the model has integer optimal solutions when the data are integer;
- write the dual with the potentials π_i, compute the reduced costs and certify the optimality of a flow;
- recognize the shortest path, the maximum flow, the transportation and the assignment problems as minimum cost flow models.

4.1 Networks and flows

Definition 4.1: network

A **network** is a directed graph $G = (N, A)$, with $n = |N|$ nodes and $m = |A|$ arcs, together with the following data.

- Every node $i \in N$ has a **balance** $b_i \in \mathbb{R}$. The node is a **supply node** if $b_i > 0$, a **demand node** if $b_i < 0$, and a **transshipment node** if $b_i = 0$.
- Every arc $(i, j) \in A$ has a **lower bound** l_{ij} and a **capacity** u_{ij}, with $0 \le l_{ij} \le u_{ij} \le +\infty$, and a **cost** $c_{ij} \in \mathbb{R}$ per unit of flow.

When the lower bounds are not given, $l_{ij} = 0$ for every arc.

Definition 4.2: flow

Let $G = (N, A)$ be a network. A **flow** is a vector $\boldsymbol{x} \in \mathbb{R}^m$ with one entry x_{ij}, the amount sent along the arc, for every $(i, j) \in A$.

- The flow satisfies the **flow conservation** constraint of node i if what leaves i minus what enters i equals the balance:

$$\sum_{(i,j)\in\delta^+(i)} x_{ij} \;-\; \sum_{(j,i)\in\delta^-(i)} x_{ji} = b_i.$$

- The flow is **feasible** if it satisfies flow conservation at every node and $l_{ij} \le x_{ij} \le u_{ij}$ for every arc. Its **cost** is $\sum_{(i,j)\in A} c_{ij}\, x_{ij}$.

- A supply node sends out b_i units more than it receives, a demand node receives $-b_i$ units more than it sends, and a transshipment node passes on everything it receives.
- With the node–arc incidence matrix $\boldsymbol{B}$ of Chapter 1 «Graphs», flow conservation at all the nodes reads $\boldsymbol{Bx} = \boldsymbol{b}$: row i of $\boldsymbol{Bx}$ is the outflow minus the inflow of i.

Example 4.1: the reference network

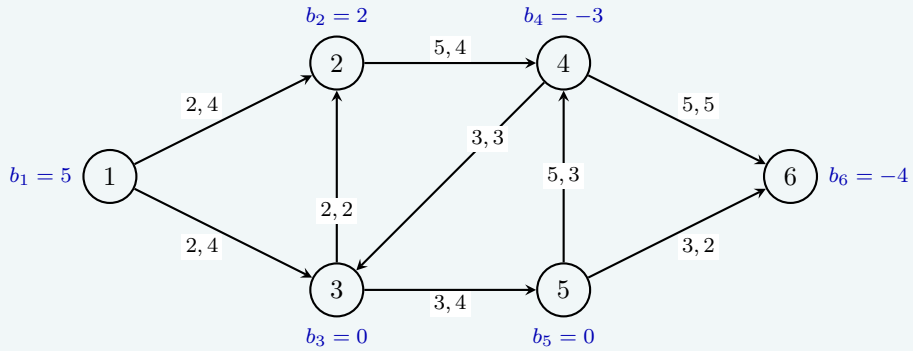
We add data to the reference graph of Chapter 1 «Graphs», with the arcs numbered $a_1, a_2, \ldots, a_9$ in lexicographic order. The balances are

$$b_1 = 5, \qquad b_2 = 2, \qquad b_3 = 0, \qquad b_4 = -3, \qquad b_5 = 0, \qquad b_6 = -4,$$

so nodes 1 and 2 supply 7 units, nodes 4 and 6 demand them, and nodes 3 and 5 are transshipment nodes. The lower bounds are zero, and the costs and capacities are:

arc	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
c_{ij}	2	2	5	2	3	3	5	5	3
u_{ij}	4	4	4	2	4	3	5	3	2

In the figure each arc carries the pair c_{ij}, u_{ij}, and each node its balance b_i (in blue).



The flow $\bar{\boldsymbol{x}}$ with

$$\bar{x}_{12} = 1,\ \bar{x}_{13} = 4,\ \bar{x}_{24} = 3,\ \bar{x}_{32} = 0,\ \bar{x}_{35} = 4,\ \bar{x}_{43} = 0,\ \bar{x}_{46} = 2,\ \bar{x}_{54} = 2,\ \bar{x}_{56} = 2$$

is feasible. For example, at node 4 the outflow is $\bar{x}_{43} + \bar{x}_{46} = 2$, the inflow is $\bar{x}_{24} + \bar{x}_{54} = 5$, and $2 - 5 = -3 = b_4$. The arcs $(1,3)$, $(3,5)$ and $(5,6)$ are at capacity, and the cost of $\bar{\boldsymbol{x}}$ is 63.

4.1.1 Flow across a cut

- For a set of nodes $S \subseteq N$, we denote by $\delta^+(S)$ the arcs with the tail in S and the head outside S, by $\delta^-(S)$ the arcs with the head in S and the tail outside S, and $x(F) = \sum_{(i,j)\in F} x_{ij}$ for a set of arcs F.

Lemma 4.3: flow across a cut

Let $\boldsymbol{x}$ satisfy flow conservation at every node of a network $G = (N, A)$. Then, for every $S \subseteq N$,

$$x(\delta^+(S)) - x(\delta^-(S)) = \sum_{i\in S} b_i.$$

Proof. Add up the conservation constraints of the nodes of S; the right-hand side is $\sum_{i\in S} b_i$.

- An arc with both endpoints in S appears twice, with $+x_{ij}$ in the row of its tail and $-x_{ij}$ in the row of its head: it cancels.
- An arc of $\delta^+(S)$ appears only in the row of its tail, with $+x_{ij}$; an arc of $\delta^-(S)$ only in the row of its head, with $-x_{ij}$; the other arcs do not appear. □

- In Example 4.1, the set $S = \{1,2,3\}$ has $\delta^+(S) = \{(2,4),(3,5)\}$ and $\delta^-(S) = \{(4,3)\}$. The flow $\bar{\boldsymbol{x}}$ gives $3 + 4 - 0 = 7 = b_1 + b_2 + b_3$.

A network has balances on the nodes and lower bounds, capacities and costs on the arcs. A feasible flow satisfies $\boldsymbol{Bx} = \boldsymbol{b}$ and the bounds; across every cut, the net outflow equals the total balance of the nodes inside.

4.2 The minimum cost flow model

Definition 4.4: minimum cost flow problem

Given a network $G = (N, A)$ with balances b_i, lower bounds l_{ij}, capacities u_{ij} and costs c_{ij}, the **minimum cost flow problem** asks for a feasible flow of minimum cost.

- We use one continuous variable x_{ij} for every arc $(i, j) \in A$, the flow on the arc.

Using these variables, a Linear Programming (LP) model for the minimum cost flow problem reads as follows:

$$\min \sum_{(i,j)\in A} c_{ij}\, x_{ij} \tag{4.1a}$$

$$\text{subject to} \sum_{(i,j)\in\delta^+(i)} x_{ij} - \sum_{(j,i)\in\delta^-(i)} x_{ji} = b_i, \qquad \forall i \in N, \tag{4.1b}$$

$$l_{ij} \le x_{ij} \le u_{ij}, \qquad \forall (i,j) \in A. \tag{4.1c}$$

This LP model is referred to as the MCF model.

- Description of the objective function and constraints of the MCF model:
  - the linear objective function (4.1a) minimizes the total cost of the flow;
  - constraints (4.1b) are the flow conservation constraints, one for every node;
  - finally, constraints (4.1c) impose the lower bound and the capacity of every arc.
- With the node–arc incidence matrix $\boldsymbol{B}$, the vector $\boldsymbol{c}$ of the costs and the vectors $\boldsymbol{l}$, $\boldsymbol{u}$ of the bounds, the MCF model is

$$\min \left\{ \boldsymbol{c}'\boldsymbol{x} \;:\; \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \le \boldsymbol{x} \le \boldsymbol{u} \right\}.$$

- The model has m variables and n equations. The column of x_{ij} in $\boldsymbol{B}$ has only two nonzero entries, $+1$ in row i and -1 in row j: this structure gives all the properties of this chapter.

Example 4.2: the MCF model of the reference network

For the network of Example 4.1 the MCF model has 9 variables and 6 equations:

$$\begin{array}{rrrrrrrrrrrr}
\min & 2x_{12} & +2x_{13} & +5x_{24} & +2x_{32} & +3x_{35} & +3x_{43} & +5x_{46} & +5x_{54} & +3x_{56} & & \\
\text{subject to} & x_{12} & +x_{13} & & & & & & & & = & 5 \\
& -x_{12} & & +x_{24} & -x_{32} & & & & & & = & 2 \\
& & -x_{13} & & +x_{32} & +x_{35} & -x_{43} & & & & = & 0 \\
& & & -x_{24} & & & +x_{43} & +x_{46} & -x_{54} & & = & -3 \\
& & & & & -x_{35} & & & +x_{54} & +x_{56} & = & 0 \\
& & & & & & & -x_{46} & & -x_{56} & = & -4
\end{array}$$

together with $0 \le x_{12} \le 4$, $0 \le x_{13} \le 4$, $0 \le x_{24} \le 4$, $0 \le x_{32} \le 2$, $0 \le x_{35} \le 4$,

$0 \le x_{43} \le 3$, $0 \le x_{46} \le 5$, $0 \le x_{54} \le 3$, $0 \le x_{56} \le 2$. The coefficient matrix of the equations is the incidence matrix $\boldsymbol{B}$ of the reference graph.

The minimum cost flow problem is the LP $\min\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \le \boldsymbol{x} \le \boldsymbol{u}\}$: one variable per arc, one conservation equation per node.

4.3 Feasibility conditions

Definition 4.5: balanced network

A network is **balanced** if $\sum_{i \in N} b_i = 0$, that is, if the total supply equals the total demand.

Proposition 4.6: a feasible network is balanced

If a network $G = (N, A)$ has a feasible flow, then $\sum_{i \in N} b_i = 0$.

Proof. Apply Lemma 4.3 with $S = N$: no arc leaves or enters N, so $\delta^+(N) = \delta^-(N) = \emptyset$ and $\sum_{i \in N} b_i = 0$. In matrix form, the rows of $\boldsymbol{B}$ add up to the zero vector, hence so do the entries of $\boldsymbol{b} = \boldsymbol{B}\boldsymbol{x}$. □

Proposition 4.7: cut condition

If a network $G = (N, A)$ has a feasible flow, then for every $S \subseteq N$

$$\sum_{i \in S} b_i \ \le\ u(\delta^+(S)) - l(\delta^-(S)),$$

where $u(F)$ and $l(F)$ are the sums of the capacities and of the lower bounds of the arcs of F.

Proof. Let $\boldsymbol{x}$ be a feasible flow. By Lemma 4.3 and the bounds $x_{ij} \le u_{ij}$ on $\delta^+(S)$ and $x_{ij} \ge l_{ij}$ on $\delta^-(S)$,

$$\sum_{i \in S} b_i = x(\delta^+(S)) - x(\delta^-(S)) \le u(\delta^+(S)) - l(\delta^-(S)).$$ □

- The supply of the nodes of S, net of their demand, must be able to leave S through the arcs of $\delta^+(S)$, while the arcs of $\delta^-(S)$ bring at least their lower bounds into S.
- A balanced network can still have no feasible flow: the balance is only a necessary condition.

Example 4.3: balanced networks without a feasible flow

We keep the arcs, costs and capacities of the reference network and change the balances.

1. With $b_5 = -1$, $b_6 = 1$ and $b_i = 0$ for the other nodes, the network is balanced. But

node 6 has no leaving arc: for $S = \{6\}$ we get $\delta^+(S) = \emptyset$, and the cut condition fails, since $b_6 = 1 > 0$.

2. With $b_1 = 9$, $b_6 = -8$ and the other balances as in Example 4.1, the network is balanced, since $9+2-3-8 = 0$. For $S = \{1\}$ we get $u(\delta^+(S)) = u_{12}+u_{13} = 8 < 9 = b_1$: node 1 cannot send out its supply.

For the reference network itself, all the 62 sets S with $\emptyset \neq S \neq N$ satisfy the cut condition, and the flow $\bar{\boldsymbol{x}}$ of Example 4.1 is feasible.

A network with a feasible flow is balanced, and every cut can carry the net supply of the nodes inside it. These conditions are necessary: a balanced network may have no feasible flow.

4.4 Removing the lower bounds

Definition 4.8: substitution of the lower bounds

Given a network with lower bounds $\boldsymbol{l}$, the **substituted network** has the same nodes, arcs and costs, zero lower bounds and

$$b'_i = b_i - \sum_{(i,j)\in\delta^+(i)} l_{ij} + \sum_{(j,i)\in\delta^-(i)} l_{ji}, \quad \forall i \in N, \qquad u'_{ij} = u_{ij} - l_{ij}, \quad \forall (i,j) \in A,$$

that is, $\boldsymbol{b}' = \boldsymbol{b} - \boldsymbol{B}\boldsymbol{l}$ and $\boldsymbol{u}' = \boldsymbol{u} - \boldsymbol{l}$.

- The l_{ij} units that must travel on (i,j) are sent in advance: the tail i has l_{ij} units less to send, the head j has l_{ij} units less to receive, and the arc has l_{ij} units less of free capacity.

Proposition 4.9: removing the lower bounds

The map $\boldsymbol{x}' = \boldsymbol{x} - \boldsymbol{l}$ is a one-to-one correspondence between the feasible flows of a network and those of its substituted network, and

$$\boldsymbol{c}'\boldsymbol{x} = \boldsymbol{c}'\boldsymbol{x}' + \boldsymbol{c}'\boldsymbol{l}.$$

Hence the two MCF models have the same optimal solutions, up to the shift by $\boldsymbol{l}$, and their optimal values differ by the constant $\boldsymbol{c}'\boldsymbol{l}$.

Proof. The map is linear and invertible, with inverse $\boldsymbol{x} = \boldsymbol{x}' + \boldsymbol{l}$. For every $\boldsymbol{x} \in \mathbb{R}^m$:

1. $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}$ if and only if $\boldsymbol{B}\boldsymbol{x}' = \boldsymbol{B}\boldsymbol{x} - \boldsymbol{B}\boldsymbol{l} = \boldsymbol{b} - \boldsymbol{B}\boldsymbol{l} = \boldsymbol{b}'$;
2. $\boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}$ if and only if $\boldsymbol{0} \leq \boldsymbol{x}' \leq \boldsymbol{u} - \boldsymbol{l} = \boldsymbol{u}'$;
3. $\boldsymbol{c}'\boldsymbol{x} = \boldsymbol{c}'(\boldsymbol{x}' + \boldsymbol{l}) = \boldsymbol{c}'\boldsymbol{x}' + \boldsymbol{c}'\boldsymbol{l}$.

The cost of every feasible flow changes by the same constant, so the optimal solutions correspond. □

Example 4.4: a lower bound on the reference network

In the network of Example 4.1 we impose $l_{32} = 1$: at least one unit must travel from 3 to 2. The column of $(3, 2)$ in $\boldsymbol{B}$ has $+1$ in row 3 and -1 in row 2, so $\boldsymbol{Bl} = (0, -1, 1, 0, 0, 0)'$ and

$$\boldsymbol{b}' = \boldsymbol{b} - \boldsymbol{Bl} = (5,\ 3,\ -1,\ -3,\ 0,\ -4)', \qquad u'_{32} = 2 - 1 = 1, \qquad \boldsymbol{c}'\boldsymbol{l} = c_{32}\, l_{32} = 2.$$

Node 3 becomes a demand node, because it has already sent one unit, and node 2 has one more unit to send. The substituted network has optimal value 60; adding the constant 2, the optimal value with $l_{32} = 1$ is 62.

The substitution $\boldsymbol{x} = \boldsymbol{l} + \boldsymbol{x}'$ removes the lower bounds: $\boldsymbol{b}' = \boldsymbol{b} - \boldsymbol{Bl}$, $\boldsymbol{u}' = \boldsymbol{u} - \boldsymbol{l}$, and the objective changes by the constant $\boldsymbol{c}'\boldsymbol{l}$.

4.5 Unbalanced instances

Definition 4.10: unbalanced instance

An instance with $\sum_{i \in N} b_i > 0$ is **unbalanced**: the total supply exceeds the total demand. Every demand must be met, and every supply node i sends out *at most* b_i units:

$$\sum_{(i,j) \in \delta^+(i)} x_{ij} - \sum_{(j,i) \in \delta^-(i)} x_{ji} \ \le\ b_i \quad \forall i \in N : b_i > 0, \qquad = b_i \quad \forall i \in N : b_i \le 0,$$

with the bounds $l_{ij} \le x_{ij} \le u_{ij}$ and the objective $\min \sum_{(i,j) \in A} c_{ij}\, x_{ij}$.

Definition 4.11: dummy node

Let $P = \{i \in N : b_i > 0\}$ be the supply nodes and $e = \sum_{i \in N} b_i > 0$ the excess. The **balanced network** adds a **dummy node** d with balance $b_d = -e$, and an arc (i, d) for every $i \in P$ with $l_{id} = 0$, $u_{id} = +\infty$ and $c_{id} = 0$.

Proposition 4.12: the dummy node

The feasible solutions of the unbalanced instance and the feasible flows of the balanced network correspond, with the same cost: a flow of the balanced network restricted to A is feasible for the unbalanced instance, and a solution of the unbalanced instance becomes a feasible flow with $x_{id} = b_i - (\text{outflow} - \text{inflow of } i)$.

Proof. For a vector $\boldsymbol{x}$ on A, let $s_i(\boldsymbol{x})$ be the outflow minus the inflow of node i along the arcs of A.

1. Let $(\boldsymbol{x}, (x_{id})_{i \in P})$ be feasible for the balanced network. At $i \in P$ conservation reads $s_i(\boldsymbol{x}) + x_{id} = b_i$ with $x_{id} \ge 0$, so $s_i(\boldsymbol{x}) \le b_i$; at the other nodes of N it reads $s_i(\boldsymbol{x}) = b_i$.

2. Conversely, let $\boldsymbol{x}$ be feasible for the unbalanced instance, and set $x_{id} = b_i - s_i(\boldsymbol{x}) \ge$

0 for $i \in P$. Conservation holds at every node of N, and at d the inflow is

$$\sum_{i \in P} x_{id} = \sum_{i \in P} b_i - \sum_{i \in N} s_i(\boldsymbol{x}) + \sum_{i \notin P} s_i(\boldsymbol{x}) = \sum_{i \in P} b_i - 0 + \sum_{i \notin P} b_i = e,$$

since the values $s_i(\boldsymbol{x})$ add up to zero: every arc of A contributes $+x_{ij}$ to its tail and $-x_{ij}$ to its head. Hence d receives exactly $-b_d = e$ units.

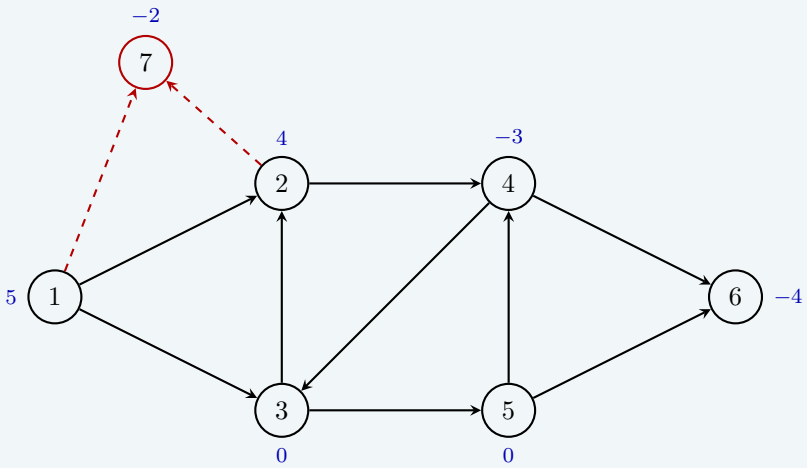
The dummy arcs cost zero, so the two costs coincide. □

- The flow on (i, d) is the part of the supply of i that is not used.
- If the total demand exceeds the total supply and the demands are upper limits, a dummy node d with balance $b_d = -\sum_{i \in N} b_i$ and arcs (d, j) to the demand nodes balances the network in the same way.

Example 4.5: a dummy node on the reference network

In the network of Example 4.1 we raise the supply of node 2 to $b_2 = 4$. The total supply is 9, the total demand is 7, and the excess is $e = 2$.

- The balanced network adds node 7 with $b_7 = -2$, and the arcs $(1, 7)$ and $(2, 7)$ with cost 0 and capacity $+\infty$ (dashed in the figure).
- Its optimal flow, computed with Gurobi, ships all the 4 units of node 2 and only 3 units of node 1: $x_{17} = 2$, $x_{27} = 0$, with optimal value 56.



An unbalanced instance becomes a balanced network with a dummy node that absorbs the excess through zero-cost arcs; the flow on a dummy arc is the unused supply of its tail.

4.6 Total unimodularity and integrality

Definition 4.13: totally unimodular matrix

An integer matrix $\boldsymbol{M}$ is **totally unimodular** (TU) if the determinant of every square submatrix of $\boldsymbol{M}$ is 0, $+1$ or -1. A square submatrix is obtained by choosing k rows and k columns, for any $k \geq 1$.

- With $k = 1$ the definition requires every entry of a TU matrix to be 0, $+1$ or -1.

Theorem 4.14: the incidence matrix is totally unimodular

The node–arc incidence matrix $\boldsymbol{B}$ of every directed graph $G = (N, A)$ is totally unimodular.

Proof. Let $\boldsymbol{M}$ be a $k \times k$ submatrix of $\boldsymbol{B}$; we prove $\det(\boldsymbol{M}) \in \{0, +1, -1\}$ by induction on k. For $k = 1$ the entry is 0, $+1$ or -1. Let $k \geq 2$. Every column of $\boldsymbol{B}$ has exactly one $+1$ and one -1, so every column of $\boldsymbol{M}$ has at most two nonzero entries; there are three cases.

1. *A column of $\boldsymbol{M}$ is zero.* Then $\det(\boldsymbol{M}) = 0$.
2. *A column of $\boldsymbol{M}$ has exactly one nonzero entry, ± 1.* The Laplace expansion along this column gives $\det(\boldsymbol{M}) = \pm \det(\boldsymbol{M}')$, where $\boldsymbol{M}'$ is a $(k-1) \times (k-1)$ submatrix of $\boldsymbol{B}$: by induction $\det(\boldsymbol{M}') \in \{0, +1, -1\}$.
3. *Every column of $\boldsymbol{M}$ has two nonzero entries, $+1$ and -1.* Then the rows of $\boldsymbol{M}$ add up to the zero vector, they are linearly dependent, and $\det(\boldsymbol{M}) = 0$. □

Example 4.6: submatrices of the reference incidence matrix

We take rows $2, 3, 4$ of the incidence matrix $\boldsymbol{B}$ of the reference graph and two choices of three columns.

	a_1	a_3	a_5
2	-1	1	0
3	0	0	1
4	0	-1	0

	a_3	a_4	a_6
2	1	-1	0
3	0	1	-1
4	-1	0	1

- The column of $a_1 = (1, 2)$ in the first matrix has one nonzero entry (case 2): expanding along it, and then along the column of a_5, the determinant is -1.
- The arcs $a_3 = (2, 4)$, $a_4 = (3, 2)$, $a_6 = (4, 3)$ of the second matrix form the cycle $2, 4, 3, 2$, and all their endpoints are among the chosen rows: every column has $+1$ and -1 (case 3), and the determinant is 0.
- The matrix $\boldsymbol{B}$ has 5004 square submatrices; computing all their determinants confirms that they take only the values -1, 0 and $+1$.

4.6.1 Integer vertices

- A point $\bar{\boldsymbol{x}}$ of a polyhedron P is a **vertex** if it is not the midpoint of two distinct points of P, as in Chapter 15 «The geometry of Linear Programming» of the module Linear Programming.
- For a set of arcs F, we denote by $\boldsymbol{B}_F$ the submatrix of the columns of F, and by $\boldsymbol{x}_F$ the corresponding entries of $\boldsymbol{x}$.

Theorem 4.15: integer vertices

Let $G = (N, A)$ be a network with integer balances $\boldsymbol{b}$, integer lower bounds $\boldsymbol{l}$ and capacities $\boldsymbol{u}$ that are integer or $+\infty$. Then every vertex of

$$P = \left\{ \boldsymbol{x} \in \mathbb{R}^m : \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \le \boldsymbol{x} \le \boldsymbol{u} \right\}$$

is an integer vector.

Proof. Let $\bar{\boldsymbol{x}}$ be a vertex of P, and let $F = \{\, a \in A : l_a < \bar{x}_a < u_a \,\}$ be the arcs strictly between their bounds. If $F = \emptyset$, every entry of $\bar{\boldsymbol{x}}$ is a bound, hence an integer.

1. *The columns of $\boldsymbol{B}_F$ are linearly independent.* Otherwise $\boldsymbol{B}_F \boldsymbol{d}_F = \boldsymbol{0}$ for some $\boldsymbol{d} \neq \boldsymbol{0}$ with $d_a = 0$ for $a \notin F$. For a small $\varepsilon > 0$ both $\bar{\boldsymbol{x}} + \varepsilon \boldsymbol{d}$ and $\bar{\boldsymbol{x}} - \varepsilon \boldsymbol{d}$ satisfy the strict bounds on F and $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}$, so they lie in P, and $\bar{\boldsymbol{x}}$ is their midpoint: a contradiction.
2. *The system for $\bar{\boldsymbol{x}}_F$ has integer data.* Every $a \notin F$ has $\bar{x}_a \in \{l_a, u_a\}$, an integer, so $\boldsymbol{B}_F \bar{\boldsymbol{x}}_F = \boldsymbol{b} - \sum_{a \notin F} \boldsymbol{B}_a \bar{x}_a =: \hat{\boldsymbol{b}}$ is integer.
3. *Cramer's rule.* $\boldsymbol{B}_F$ has rank $|F|$, so some $|F|$ of its rows form a nonsingular square matrix $\boldsymbol{M}$, and $\boldsymbol{M}\bar{\boldsymbol{x}}_F = \hat{\boldsymbol{b}}_R$ on these rows R. By Cramer's rule each entry of $\bar{\boldsymbol{x}}_F$ is $\det(\boldsymbol{M}_a)/\det(\boldsymbol{M})$, with $\boldsymbol{M}_a$ an integer matrix.
4. By Theorem 4.14, $\det(\boldsymbol{M}) = \pm 1$, so every entry of $\bar{\boldsymbol{x}}_F$ is an integer. □

Corollary 4.16: integrality of the minimum cost flow model

Let the balances, the lower bounds and the capacities be integer, with every capacity finite. If the MCF model has a feasible solution, then it has an integer optimal solution, for any costs $\boldsymbol{c}$.

Proof. The polyhedron P is nonempty and bounded, since $\boldsymbol{l} \le \boldsymbol{x} \le \boldsymbol{u}$ with finite bounds. A linear function attains its minimum over a nonempty bounded polyhedron at a vertex (Chapter 15 «The geometry of Linear Programming»), and every vertex of P is integer by Theorem 4.15. □

- The simplex algorithm ends at a vertex, so with integer data it returns an integer optimal flow: the integer requirement $x_{ij} \in \mathbb{Z}_{\ge 0}$ comes for free.

- Total unimodularity is a property of the matrix: for a matrix that is not TU, an integer right-hand side can give fractional vertices.

Example 4.7: a matrix that is not totally unimodular

The undirected triangle with vertices $1, 2, 3$ and edges $e_1 = \{1, 2\}$, $e_2 = \{1, 3\}$, $e_3 = \{2, 3\}$ has the vertex–edge incidence matrix

$$\boldsymbol{M} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \det(\boldsymbol{M}) = -2,$$

so $\boldsymbol{M}$ is not TU. The polyhedron $\{\boldsymbol{y} \in \mathbb{R}^3 : \boldsymbol{M}\boldsymbol{y} = \boldsymbol{1},\ \boldsymbol{y} \geq \boldsymbol{0}\}$ contains the single point $\boldsymbol{y} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, since $\boldsymbol{M}$ is nonsingular: this vertex is fractional, although the data are integer.

The incidence matrix of a directed graph is totally unimodular. Hence, with integer balances and bounds, every vertex of the feasible region is integer, and the MCF model has an integer optimal solution.

4.7 The dual model and the reduced costs

- By Proposition 4.9 the lower bounds can be removed, so in this section $\boldsymbol{l} = \boldsymbol{0}$ and every capacity is finite. The constraint $x_{ij} \leq u_{ij}$ is written as $-x_{ij} \geq -u_{ij}$.
- Following the rules of Chapter 9 «Duality theory» of the module Linear Programming, we use a free dual variable π_i, the **potential** of node i, for its conservation equation, and a non-negative dual variable α_{ij} for the capacity of (i, j).

Using free dual variables π_i for the constraints (4.1b) and non-negative dual variables α_{ij} for the capacities, the dual model of the MCF model with $\boldsymbol{l} = \boldsymbol{0}$ is:

$$\max \sum_{i \in N} b_i \, \pi_i - \sum_{(i,j) \in A} u_{ij} \, \alpha_{ij} \tag{4.2a}$$

$$\text{subject to} \quad \pi_i - \pi_j - \alpha_{ij} \leq c_{ij}, \qquad \forall (i, j) \in A, \tag{4.2b}$$

$$\pi_i \in \mathbb{R}, \qquad \forall i \in N, \tag{4.2c}$$

$$\alpha_{ij} \geq 0, \qquad \forall (i, j) \in A. \tag{4.2d}$$

This LP model is referred to as the D(MCF) model.

- Description of the dual model:
  - the column of x_{ij} has $+1$ in the equation of i, -1 in the equation of j and -1 in its capacity constraint, which gives the dual constraint (4.2b), one for every arc;
  - the primal variables are non-negative, so the dual constraints are of type $\leq$.

- Adding the same constant to all the potentials changes neither (4.2b) nor the objective, since $\sum_{i \in N} b_i = 0$: the potentials can be normalized, for example with $\pi_n = 0$.

Definition 4.17: reduced cost

Given potentials $\boldsymbol{\pi} \in \mathbb{R}^n$, the **reduced cost** of the arc $(i,j) \in A$ is

$$\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j.$$

- The dual constraint (4.2b) reads $\bar{c}_{ij} + \alpha_{ij} \geq 0$. For fixed potentials, the best choice of $\alpha_{ij} \geq 0$ is the smallest one, $\alpha_{ij} = \max\{0, -\bar{c}_{ij}\}$.

Lemma 4.18: cost and reduced cost

For every $\boldsymbol{x}$ with $\boldsymbol{Bx} = \boldsymbol{b}$ and every $\boldsymbol{\pi} \in \mathbb{R}^n$,

$$\sum_{(i,j) \in A} c_{ij}\, x_{ij} = \sum_{(i,j) \in A} \bar{c}_{ij}\, x_{ij} + \sum_{i \in N} b_i\, \pi_i.$$

Proof. Since $c_{ij} = \bar{c}_{ij} + \pi_i - \pi_j$, it suffices to show $\sum_{(i,j) \in A} (\pi_i - \pi_j)\, x_{ij} = \sum_{i \in N} b_i \pi_i$. The term $\pi_i x_{ij}$ goes to the tail and $-\pi_j x_{ij}$ to the head, so

$$\sum_{(i,j) \in A} (\pi_i - \pi_j)\, x_{ij} = \sum_{i \in N} \pi_i \Big(\sum_{(i,j) \in \delta^+(i)} x_{ij} - \sum_{(j,i) \in \delta^-(i)} x_{ji} \Big) = \sum_{i \in N} \pi_i\, b_i. \qquad \square$$

Theorem 4.19: duality gap

Let $\boldsymbol{x}$ be a feasible flow ($\boldsymbol{l} = \boldsymbol{0}$) and $(\boldsymbol{\pi}, \boldsymbol{\alpha})$ a feasible solution of the D(MCF) model. Then

$$\sum_{(i,j) \in A} c_{ij} x_{ij} - \Big(\sum_{i \in N} b_i \pi_i - \sum_{(i,j) \in A} u_{ij} \alpha_{ij} \Big) = \sum_{(i,j) \in A} (\bar{c}_{ij} + \alpha_{ij})\, x_{ij} + \sum_{(i,j) \in A} \alpha_{ij}\, (u_{ij} - x_{ij}) \;\geq\; 0,$$

and both solutions are optimal if and only if, for every arc, $(\bar{c}_{ij} + \alpha_{ij})\, x_{ij} = 0$ and $\alpha_{ij}(u_{ij} - x_{ij}) = 0$.

Proof. By Lemma 4.18, the left-hand side is $\sum \bar{c}_{ij} x_{ij} + \sum u_{ij} \alpha_{ij}$, and $u_{ij}\alpha_{ij} = \alpha_{ij} x_{ij} + \alpha_{ij}(u_{ij} - x_{ij})$ gives the identity.

- Every term is non-negative: $\bar{c}_{ij} + \alpha_{ij} \geq 0$ by (4.2b), $\alpha_{ij} \geq 0$, and $0 \leq x_{ij} \leq u_{ij}$.
- If the gap is zero, the primal cost equals a dual value, which is a lower bound on every primal cost by the inequality just proved: $\boldsymbol{x}$ is optimal, and in the same way $(\boldsymbol{\pi}, \boldsymbol{\alpha})$ is optimal.
- If both are optimal, the gap is zero by the strong duality theorem of Chapter 9 «Duality theory». A sum of non-negative terms is zero if and only if every term is zero. $\square$

Corollary 4.20: reduced cost certificate

Let $\boldsymbol{x}$ be a feasible flow ($\boldsymbol{l} = \mathbf{0}$) and $\boldsymbol{\pi} \in \mathbb{R}^n$ potentials such that, for every arc,

$$\bar{c}_{ij} > 0 \;\Rightarrow\; x_{ij} = 0, \qquad \bar{c}_{ij} < 0 \;\Rightarrow\; x_{ij} = u_{ij}.$$

Then $\boldsymbol{x}$ is an optimal flow, and $\boldsymbol{\pi}$ with $\alpha_{ij} = \max\{0, -\bar{c}_{ij}\}$ is an optimal solution of the D(MCF) model.

Proof. The pair $(\boldsymbol{\pi}, \boldsymbol{\alpha})$ is dual feasible, since $\alpha_{ij} \geq 0$ and $\bar{c}_{ij} + \alpha_{ij} \geq 0$. We check the two conditions of Theorem 4.19 on every arc.

- If $\bar{c}_{ij} \geq 0$, then $\alpha_{ij} = 0$, and $\bar{c}_{ij} x_{ij} = 0$ because $\bar{c}_{ij} > 0$ forces $x_{ij} = 0$.
- If $\bar{c}_{ij} < 0$, then $\bar{c}_{ij} + \alpha_{ij} = 0$, and $\alpha_{ij}(u_{ij} - x_{ij}) = 0$ because $x_{ij} = u_{ij}$.

The gap is zero, so both solutions are optimal. □

- An arc with positive reduced cost is empty, an arc with negative reduced cost is saturated, and an arc strictly between its bounds has reduced cost zero.

Example 4.8: the dual of the reference network

For the network of Example 4.1, the D(MCF) model is

$$\max\; 5\pi_1 + 2\pi_2 - 3\pi_4 - 4\pi_6 - \left(4\alpha_{12} + 4\alpha_{13} + 4\alpha_{24} + 2\alpha_{32} + 4\alpha_{35} + 3\alpha_{43} + 5\alpha_{46} + 3\alpha_{54} + 2\alpha_{56}\right)$$

with $\boldsymbol{\alpha} \geq \mathbf{0}$, free potentials, and one constraint per arc:

$$\begin{aligned} &\pi_1 - \pi_2 - \alpha_{12} \leq 2, \quad \pi_1 - \pi_3 - \alpha_{13} \leq 2, \quad \pi_2 - \pi_4 - \alpha_{24} \leq 5, \\ &\pi_3 - \pi_2 - \alpha_{32} \leq 2, \quad \pi_3 - \pi_5 - \alpha_{35} \leq 3, \quad \pi_4 - \pi_3 - \alpha_{43} \leq 3, \\ &\pi_4 - \pi_6 - \alpha_{46} \leq 5, \quad \pi_5 - \pi_4 - \alpha_{54} \leq 5, \quad \pi_5 - \pi_6 - \alpha_{56} \leq 3. \end{aligned}$$

The balances $b_3 = b_5 = 0$ make π_3 and π_5 disappear from the objective.

- The potentials and the reduced costs are the basis of the optimality tests of Chapter 6 «Minimum cost flow».

The dual of the MCF model has a potential π_i per node; the reduced cost is $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$. A feasible flow whose positive-reduced-cost arcs are empty and negative-reduced-cost arcs are saturated is optimal.

4.8 Special cases

- Several classical problems are MCF models with special data; all the lower bounds are zero.

problem	the MCF model
Shortest path	$b_s = 1$, $b_t = -1$, $b_i = 0$ otherwise; $u_{ij} = +\infty$, c_{ij} the length of the arc: the unit of flow travels from s to t.
Maximum flow	$b_i = 0$ and $c_{ij} = 0$ for every arc, plus the arc (t, s) with $c_{ts} = -1$ and $u_{ts} = +\infty$: the flow on (t, s) is the value sent from s to t.
Transportation	$N = P \cup D$ with arcs from the origins P to the destinations D; b_i the supply of $i \in P$, $-b_j$ the demand of $j \in D$, $u_{ij} = +\infty$.
Assignment	a transportation model with $\lvert P\rvert = \lvert D\rvert$, $b_i = 1$ for $i \in P$ and $b_j = -1$ for $j \in D$.

- In the four models the balances and the bounds are integer, so the integer vertices of Theorem 4.15 are paths, integer flows and assignments.

Shortest path, maximum flow, transportation and assignment are minimum cost flow models with special balances, capacities and costs.

4.9 A transshipment instance

- We solve the MCF model of Example 4.2: the reference network, with two supply nodes, two demand nodes and two transshipment nodes. The optimal solution is computed with Gurobi (gurobipy), and then certified by hand with Corollary 4.20.

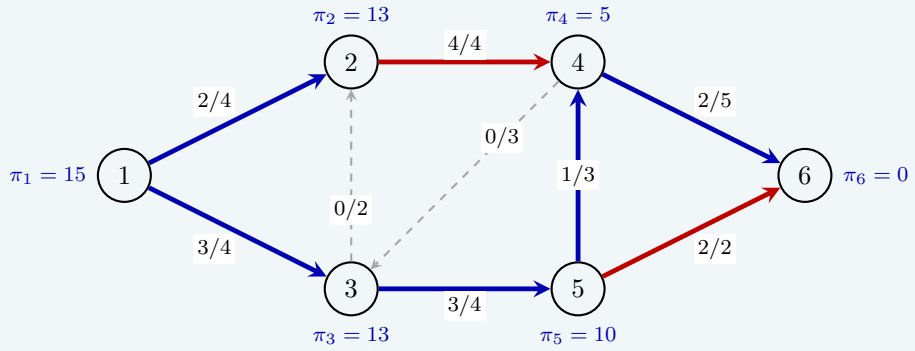
Example 4.9: the optimal flow

The optimal flow $\tilde{\boldsymbol{x}}$ and its optimal value are

arc	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
$\tilde{x}_{ij}$	2	3	4	0	3	0	2	1	2
u_{ij}	4	4	4	2	4	3	5	3	2

$z(\text{MCF}) = 60.$

In the figure each arc carries $\tilde{x}_{ij}/u_{ij}$: the arcs strictly between their bounds are blue, the arcs at capacity red, the empty arcs grey and dashed. Next to each node, its potential.



The flow is integer, as Corollary 4.16 guarantees.

4.9.1 The certificate of optimality

1. The five blue arcs $(1,2)$, $(1,3)$, $(3,5)$, $(4,6)$, $(5,4)$ form a spanning tree. We ask for reduced cost zero on them, $\pi_i - \pi_j = c_{ij}$, and normalize with $\pi_6 = 0$.
2. Going backwards from node 6: $\pi_4 = \pi_6 + c_{46} = 5$, $\pi_5 = \pi_4 + c_{54} = 10$, $\pi_3 = \pi_5 + c_{35} = 13$, $\pi_1 = \pi_3 + c_{13} = 15$ and $\pi_2 = \pi_1 - c_{12} = 13$.
3. The potential π_i is the cost of the tree path from i to 6: for example, $\pi_1 = c_{13} + c_{35} + c_{54} + c_{46} = 2 + 3 + 5 + 5 = 15$.

Example 4.10: reduced costs and optimality

The reduced costs $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$ are:

arc	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
$\bar{c}_{ij}$	0	0	-3	2	0	11	0	0	-7
α_{ij}	0	0	3	0	0	0	0	0	7

- The arcs with $\bar{c}_{ij} > 0$, $(3,2)$ and $(4,3)$, are empty; the arcs with $\bar{c}_{ij} < 0$, $(2,4)$ and $(5,6)$, are at capacity. By Corollary 4.20, $\tilde{\boldsymbol{x}}$ is optimal.
- The dual value is $\sum_{i \in N} b_i \pi_i - \sum_{(i,j) \in A} u_{ij} \alpha_{ij} = 86 - 26 = 60$, equal to the cost of $\tilde{\boldsymbol{x}}$. The potentials coincide with the dual values returned by Gurobi, shifted so that $\pi_6 = 0$.

- **Uniqueness.** Every optimal flow has gap zero with $(\boldsymbol{\pi}, \boldsymbol{\alpha})$, so by Theorem 4.19 it has $x_{32} = x_{43} = 0$, $x_{24} = 4$ and $x_{56} = 2$. The flow on the five tree arcs is then fixed by conservation, since their columns are linearly independent: $\tilde{\boldsymbol{x}}$ is the unique optimal flow.
- **The flow $\bar{\boldsymbol{x}}$.** The feasible flow $\bar{\boldsymbol{x}}$ of Example 4.1 costs 63, and its gap with the same dual solution is $63 - 60 = 3 = \alpha_{24}(u_{24} - \bar{x}_{24}) = 3 \cdot (4 - 3)$: the arc $(2,4)$ is not used enough.

On the reference network the optimal flow costs 60. The potentials of its spanning tree give reduced costs that are positive only on empty arcs and negative only on saturated arcs: they certify the optimality without a solver.

4.10 Typical mistakes

Warning

- **Swapping outflow and inflow.** The conservation constraint is outflow minus inflow equal to b_i, with $b_i > 0$ at a supply node: the signs follow the incidence matrix, $+1$ in the tail.
- **Taking $\sum_i b_i = 0$ as sufficient.** A balanced network can have no feasible flow: a cut may be too small to carry the supply inside it.
- **Getting the substitution signs wrong.** With $\boldsymbol{x} = \boldsymbol{l} + \boldsymbol{x}'$ the tail of (i, j) loses l_{ij} units of balance and the head gains them; the constant $\boldsymbol{c}'\boldsymbol{l}$ must be added back to the value.
- **Forgetting the dummy node.** An instance with more supply than demand has no feasible flow with equalities; the dummy arcs must cost zero, or the unused supply is charged.
- **Claiming integrality for any matrix.** Integer vertices need a TU matrix and integer $\boldsymbol{b}$, $\boldsymbol{l}$, $\boldsymbol{u}$; the costs may be fractional.
- **Writing the reduced cost with the wrong signs.** In this module $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$, with $-\pi$ of the tail and $+\pi$ of the head.
- **Expecting unique potentials.** The potentials are defined up to an additive constant; they are fixed by a normalization such as $\pi_n = 0$.

4.11 In summary

Summary

- A network has balances b_i, lower bounds l_{ij}, capacities u_{ij} and costs c_{ij}; the minimum cost flow problem is the LP $\min\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \le \boldsymbol{x} \le \boldsymbol{u}\}$.
- Across every cut the net outflow equals the balance inside; a feasible network is balanced and satisfies the cut condition.
- Lower bounds are removed with $\boldsymbol{x} = \boldsymbol{l} + \boldsymbol{x}'$; an excess of supply goes to a dummy node through zero-cost arcs.
- The incidence matrix is totally unimodular, so with integer $\boldsymbol{b}$, $\boldsymbol{l}$, $\boldsymbol{u}$ every vertex is integer and the MCF model has an integer optimal solution.
- The dual has potentials π_i; with $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$, a flow with empty positive-cost arcs and saturated negative-cost arcs is optimal.

- Shortest path, maximum flow, transportation and assignment are MCF models.

The reference network at a glance.

balances	$\boldsymbol{b} = (5, 2, 0, -3, 0, -4)$
optimal flow	$\tilde{\boldsymbol{x}} = (2, 3, 4, 0, 3, 0, 2, 1, 2)$, $z(\text{MCF}) = 60$
potentials ($\pi_6 = 0$)	$\boldsymbol{\pi} = (15, 13, 13, 5, 10, 0)$
empty / saturated arcs	$(3, 2)$, $(4, 3)$ / $(2, 4)$, $(5, 6)$
with $l_{32} = 1$	$\boldsymbol{b}' = (5, 3, -1, -3, 0, -4)$, optimal value 62
with $b_2 = 4$ (dummy node 7)	$x_{17} = 2$, optimal value 56